Anatomy of the Lower Face and Botulinum Toxin Injections

Patrick Trévidic, MD
Jonathan Sykes, MD
Gisella Criollo-Lamilla, MD

Paris, France; and Sacramento, Calif.

Summary: The use of botulinum toxin in the lower face is more complex and less reliable than its use in the upper face. Use in the lower face is also fraught with more adverse events. The anatomy of the lower face is complex, as the muscles of this region are very close together and interface at different levels and depths to perform heterogeneous functions, such as talking, eating, drinking, and expressivity of the face. This article, based on the anatomical knowledge of each muscle of the lower face, provides recommendations and guidelines to perform botulinum toxin injection safely. The review of each muscle with its relevant anatomy and relationship of this three-dimensional anatomy with the cutaneous plane gives the exact position of injectable locations. For each muscle, the number of points related to the motor end plate location and the exact dose related to the muscular mass are indicated. Summary tables are provided. (*Plast. Reconstr. Surg.* 136: 84S, 2015.)

The use of botulinum toxin in the lower face is more complex and less reliable than the use of botulinum toxin in the upper face. Lower face toxin use is also fraught with more adverse events.[1]

The anatomy of the lower face is complex as the muscles of this region are very close together and interface at different levels and depths to perform heterogeneous functions, such as talking, eating, drinking, or expressivity of the face.[2,3]

A special attention is paid to what we call 3D muscles, such as the levator labii superioris alaeque nasi (LLSAN), depressor anguli oris (DAO), and mentalis. The 3D muscles have various depths on their pathway: deep at the bone origin and superficial at the skin insertion point. Thus, this concept will help readers place the inferior and superior injections points in the appropriate plane.

The spreading of toxin into untargeted muscles may produce unwanted effects, such as asymmetrical facial movement, drooling, or difficulty to speak or smile. In an attempt to avoid unwanted toxin affects, the toxin is reconstituted with small volumes of diluent. Careful slow injection technique is also recommended to avoid injuring vascular pedicles. To avoid complications in the lower face, a detailed and thorough knowledge of the anatomical landmarks and danger zones is necessary. The lower face muscles, the injection points, and the related doses are schematically illustrated at the end of the article to help readers visualize and understand the pertinent lower face anatomy. In addition, recommended injection techniques and key points for each muscle are summarized.

From the Expert2Expert Group and University of California, Davis Medical Center.
Received for publication May 12, 2015; accepted July 23, 2015.
Copyright © 2015 by the American Society of Plastic Surgeons
DOI: 10.1097/PRS.0000000000001787

MUSCLES

It is essential to have a complete awareness of the muscles and their actions in this region of the face, including the wrinkles created by their movements and their interactions with neighboring structures.[4] Some muscles are called "3D muscles" because they have a deep origin and a superficial insertion, which means that they have different depths during their course. This is a very important notion because it will have repercussions on the depth of the injection at determined places.[5]

LLSAN Muscle

This is a long and narrow muscle that runs vertically from its upper origin in the frontal process of the maxilla bone, to its lower cutaneous

Disclosure: *None of the authors has a financial interest in any of the products, devices, or drugs mentioned in this article.*

Copyright © 2015 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

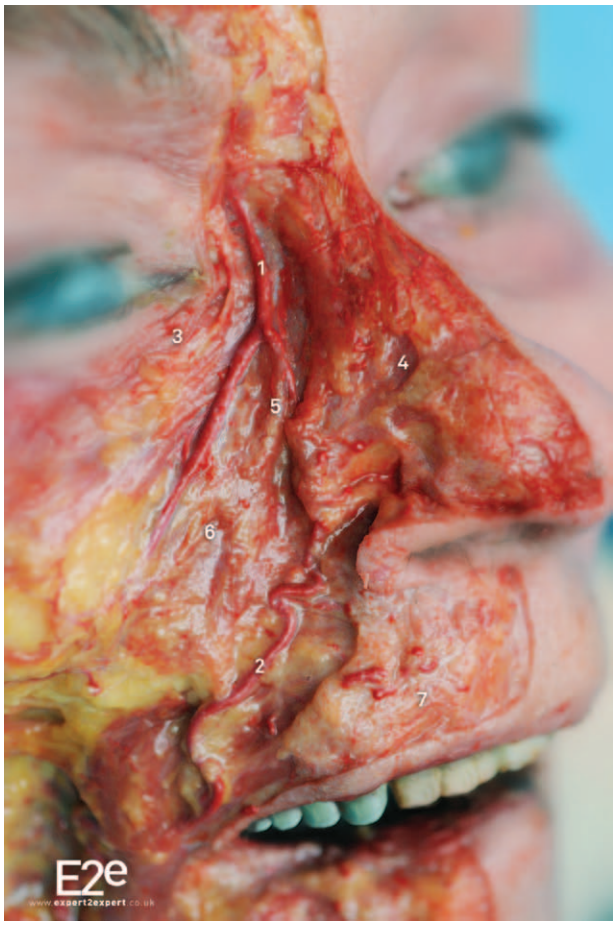


Fig. 1. View of the anatomy of the nose and mid cheek region. 1, Angular artery; 2, facial artery; 3, orbicularis oculi muscle; 4, nasalis pars transversa muscle; 5, LLSAN muscle; 6, levator labii superioris muscle; 7, orbicularis oris muscle. Figure courtesy of Expert2Expert-E2e.

insertion into the lateral nostril region (nasal portion), into the upper part of the nasolabial fold, and into the upper lip (labial portion). This muscle is considered a 3D muscle (Fig. 1).

The LLSAN superior muscular fibers are covered by the orbicularis oculi muscle. The facial vessels run along the nasojugal fold, between the orbicularis oculi and the LLSAN.[6,7] Its inferior fibers interface the orbicularis oris muscle, and its lateral fibers interdigitate with the levator labii superioris (LLS) muscle.

The nasal portion of the LLSAN produces vertical elevation and dilatation of the nostril. The labial portion is responsible for hollowing out of the upper third of the nasolabial fold and raising of the upper lip. The classical description[8,9] is that the nasalis pars transversa muscle is responsible for the downward medial oblique wrinkles on both sides of the nose called "bunny lines." However, these creases are made from the contraction of the LLSAN.[5,10] This muscle is also the primary muscle responsible for the gummy smile.[11]

Technique of injection: Only 1 injection point is made per side. Two techniques are possible: superior and deep or inferior and superficial.[5]

1. Superior and deep: To locate this injection point, first draw the inferior orbital margin. Then, draw the midpupillary line representing the emergence of the infraorbital nerve through the foramen, which is covered by the LLS. The LLSAN is located just medially to the LLS, and the injection point is placed near to its osseous origin. The tip of the needle penetrates about 3 mm deep until making contact with the bone.

2. Inferior and superficial: This point is located lateral to the nostril above the nasolabial fold to avoid any interference with the orbicularis oris muscle. The injection must be superficial in the subdermal plane. The injection point should not be too inferior, to prevent diffusion of the toxin into the orbicularis oris, resulting in an asymmetric smile.

Dose: The efficient dose is related to the total muscular mass. The recommended dosage ranges from 5 international units (IU) or 12.5 Speywood units (SU) per side (Table 1).

Orbicularis Oris Muscle

The orbicularis oris muscle is a broad elliptical muscle composed of concentric fibers forming a complete ring around the mouth[12] (Fig. 2). This muscle produces the closure and forward protrusion of the lips.[5]

It is generally accepted that the orbicularis oris is not circular but consists of 2 parts, a lower and an upper part joined to the modiolus. These 2 parts are themselves composed of 2 distinct portions, the pars marginalis and the pars peripheralis, which differ in location and function. Their fibers come from the modiolus but are directed differently and are not within the same plane.[2,13–15]

1. Pars marginalis: It extends, as an arc, under the free border of the lips, within the 2 corners of the mouth where the fibers of the superior and inferior portions blend. As the pars marginalis is located in the vermilion, it occupies a more anterior frontal plane that does the pars peripheralis, and it gives its

Copyright © 2015 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

Table 1. Botulinum Toxin Units

Generic Name	Units
OnabotulinumtoxinA	International units (IU)
IncobotulinumtoxinA	International units (IU)
AbobotulinumtoxinA	Speywood units (SU)



Fig. 2. View of the lower third of the face. 1, Orbicularis oris muscle; 2, DAO muscle; 3, DLI muscle; 4, mentalis muscle; 5, platysma muscle; 6, zygomaticus major muscle; 7, LLSAN muscle; 8, subalar artery; 9, septal artery; 10, facial artery. Figure courtesy of Expert2Expert-E2e.

curved shape at the lip. The pars marginalis portion of the muscle acts as a sphincter.

2. Pars peripheralis: Is located in the cutaneous lip; it is the thinner and more peripheral portion of the muscle. It sits in the most lateral part of the corner of the mouth and partially above the perioral muscles. The most peripheral fibers are connected with the maxillary bone and the septum of the nose above and with the mandible below. It has a dilatory function. The contraction of this portion of the muscle is responsible for accentuation of vertical fine perioral wrinkles.

Technique of injection: Two injection points are made in the superior lip and 2 in the inferior lip.

The injection has to remain quite superficial, just subcutaneous; the needle penetrates for a distance of only 2 mm until it reaches the orbicularis oris insertion in the wet-dry border. This is the only spot where the muscle comes into contact with the deep dermis of the lip (the so-called hockey stick shape of the muscle). The orbicularis oris insertion defines the wet-dry vermilion border.[12]

The injection is placed in the vermilion border in correspondence with the fine rhytids, in the middle part of each half lip because the motor endplates have a main concentration in the center of the right and left orbicularis oris quadrants (about halfway between the corner of the mouth and the facial midline).[16]

1. Avoid injecting laterally close to the corner of the mouth to prevent the toxin from diffusing into the dense interlacing fibers of the peribuccal muscles. Lateral diffusion can produce a drooping of the corners of the mouth and consequently drooling, impaired lip function, and difficulty when drinking. This is the reason why it is important to stay medially in the lip.
2. In the upper lip, avoid injecting into the midline area to prevent the risk of flattening of the philtral crest. Also, avoid treating the Cupid's bow, which could produce a flattening of the convexity of the prominence.

Botulinum toxin injection into the orbicularis oris will produce a reduction of fine vertical wrinkles that are caused by muscle contraction, but have no effect on secondary wrinkles due to dermal and epidermal aging but mainly due to smoking habits. Botulinum toxin also produces a slight eversion of the upper lip vermilion border.

Dose: 1 IU or 2.5 SU per point at a very high concentration (to minimize diffusion of toxin into surrounding tissues).

DAO Muscle

This is a triangularly shaped muscle, with a large deep base on the mandibular origin. There are many anatomical variations in the size of this muscle. The DAO runs vertically from its mandibular deep origin to its superficial insertion into the modiolus (3D muscle). This muscle covers partly the depressor labii inferioris (DLI) muscle and its osseous insertion[5] (Fig. 3).

The marionette fold creating a sad-looking mouth is caused by contraction of DAO muscle, which pulls the labial commissures inferiorly.[17–20]

The relationship between the position of this fold and the DAO is in the middle of the upper part of the body of the DAO and not at the anterior part of this muscle (Fig. 4).

Technique of injection: 2 inferior injection points and/or 1 superior are made.[5]

Copyright © 2015 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

1. Inferior injection point: Deep in the bone origin of the muscle. The mandibular ligament is located immediately behind the DAO in front of the jowl, making it a useful anatomical landmark. The 2 inferior injection points are therefore positioned anterior to this ligament, 3–5 mm above the mandibular border and 2–3 mm apart. The tip of the needle goes deep 3–4 mm until making contact with the bone at the muscle origin. Avoid injecting too medially in the chin; the diffusion to the DLI muscle can produce speaking difficulties and lip asymmetry at rest or in motion.
2. Superior injection point: Superficial in the upper midline mass of the muscle always behind the marionette lines (about 0.5 mm below and outside the labial commissure). The tip of the needle goes deep 2–3 mm in the skin and in the subcutaneous plane. It should be kept superficial. Avoid injecting too close to the corner of the lips; the diffusion to the orbicularis oris muscle can produce oral incompetence and drooling.[9]

Dose: 2 IU or 5 SU per point.



Fig. 3. View of the DAO region. 1, DAO muscle; 2, DLI muscle; 3, orbicularis oris muscle; 4, platysma muscle; 5, masseter muscle; 6, facial artery; 7, parotid duct; 8, mandibular branch of the facial nerve. Figure courtesy of Expert2Expert-E2e.

Mentalis Muscle

The fibers of this flat muscle go vertically from the superior deep origin in the mandible to the inferior superficially cutaneous insertion in the medial chin (3D muscle[5]). The mentalis muscle fibers are oriented medially downward creating with the contralateral muscle a central V-shaped triangle which contains only deep fat.[6] Laterally, the mentalis muscle connects with the DLI muscle (Fig. 2).

The mentalis muscle raises and protrudes the lower lip. Additionally, this muscle causes adherence of the skin of the chin to the subcutaneous tissue creating a dimpling effect. It is responsible for chin wrinkles and an "orange peel" dimpling aspect on the skin of the chin. It flattens the chin and gives it a pouting effect.

The motor endplates of the mentalis muscle can be found in both superior and inferior muscle portions, with a predominance of the superior location.[16]

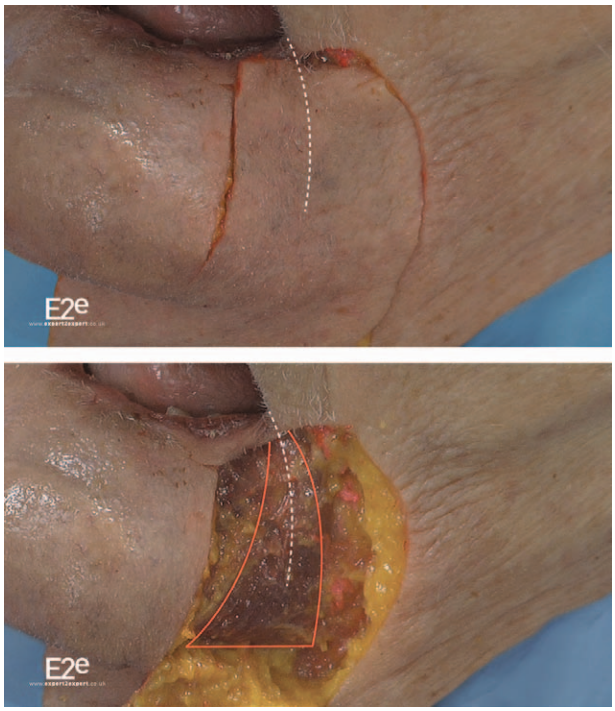


Fig. 4. View of the DAO region. The marionette line is marked by a white dotted line. It is located in the middle of the upper part of the DAO muscle body marked by a red line. Figure courtesy of Expert2Expert-E2e.

Copyright © 2015 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

Technique of injection: 1 injection point per side is made.[5,9] The injection is placed into each mentalis muscle: 1 cm above the tip of the chin and 5 mm laterally to the middle line, just below the dimpled area. The tip of the needle goes superficially 2–3 mm until reaching the superficial skin insertion.

Botulinum toxin injections into this muscle will attenuate the dimple, improve the flat appearance of the chin, relax and soften the chin skin, and reduce the labiomental fold.

Dose: 3 IU or 7.5 SU per side.

Masseter Muscle

The masseter muscle is a rectangular and thick muscle that extends from the zygomatic arch and the zygomatic bone to the ramus, mandibular angle, and coronoid process of the mandible (Figs. 3 and 5).

The masseter muscle is composed of 2 parts, pars superficialis and pars profunda; it is covered by a strong aponeurosis.[21,22] The masseter muscle belongs to the masticatory muscles.

The pars superficialis arises by a thick tendinous aponeurosis from the maxillary process of zygomatic bone and the anterior three fourths of the inferior border of the zygomatic arch. Its fibers pass backward and downward, to be inserted into the angle of the mandible and inferior part of the lateral part of the ramus of the mandible.

The pars profunda arises from the medial part of the zygomatic arch, from the posterior one fourth of the inferior border of the zygomatic arch, and from the deep layer of the deep temporal fascia. The fibers pass downward and medially to be inserted on the lateral part of the coronoid process close to the insertion of the temporalis muscle and into the superior part of the lateral part of the ramus of the mandible.



Fig. 5. View of the lower face and neck. 1, Platysma muscle; 2, sternocleidomastoideus muscle; 3, masseter muscle; 4, superficial cervical plexus; 5, masseteric ligament; 6, mandibular ligament; 7, anterior platysma band; 8, superficial fat layer. Figure courtesy of Expert2Expert-E2e.

The main connections of the masseter muscle are anterior and inferior: the facial pedicle and the platysma muscle (sometimes the posterior border of the platysma covers the inferior and anterior part of the masseter); posteriorly: the parotid gland.

Technique of injection: 3 points of injection are made in a triangle shape: 2 inferior points spaced about 1.5 cm each, following the outline of an imaginary parallel line 0.5 cm above to the inferior border of the body of the mandible. The third point, the apex of the triangle, is 2 cm above the previous points into the mass of the muscle. A long needle is used to allow contact with the bone during injection.

The motor endplates of the masseter muscle are located in the center of the lower half of this muscle.[23,24]

Botulinum toxin injections into this muscle will decrease the bruxism, reduce the hypertrophic appearance of the muscle, and reduce the size of the mandibular angle.

Dose: 8–40 IU or 20–100 SU per side.

Platysma Muscle

The platysma muscle is a flat, broad, and thin superficial muscle that runs under the thin subcutaneous tissue on each side of the neck. Its fibers extend obliquely upward and inward from the upper anterior thorax and deltoideus muscle to the mandible and to the oral commissure (Fig. 5).

The main connections of the platysma muscle are (1) superior and medially, with the lateral part of the orbicularis oris muscle, and (2) medially in the mandibular region, the anterosuperior fibers blend with the posterior border of the DAO muscle.

The platysma, together with the DAO, pulls the buccal commissures laterally downward.[25]

The anterior platysma border produces the cervical platysma bands.[26] At rest, platysma bands are due to hyperactivity of this muscle and not to a drooping effect. It is commonly thought that the platysma muscle appearance is dictated by cervical skin. In actuality, the skin of the neck follows the location and action of the platysma muscle.[5]

Technique of injection: 8–10 injection points per side are made. The injections are placed in vertical series of points in the anterior border of the muscle (into the platysma band),

Copyright © 2015 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

Table 2. Botulinum Toxin Injections of the Lower Face

Muscles	Site of Injection	Depth of Injection	No. Injection Points per Side	No. Units per Point	Comments
LLSAN	Medially to the infraorbital foramen	Deep	1	5 IU/12.5 SU	The anatomical landmark is the infraorbital nerve
	OR				
	Laterally to the nostril above the nasolabial fold	Superficial			Avoid injecting into the orbicularis oris muscle
Orbicularis oris	At the vermilion border (junction between the red and the white lips), in the mid third of each half lip	Superficial	1	1 IU/2.5 SU	Avoid injecting too laterally (close to the corner of the mouth) and too medially (close to the midline of the upper lip)
DAO	Medially to the mandibular ligament	Deep	2	2 IU/5 SU	Avoid injecting into the DLI muscle
	AND/OR				
	In the upper part of the muscle body, slightly posteriorly to the marionette line	Superficial	1		Avoid injecting into the orbicularis oris muscle
Mentalis	1 cm above the tip of the chin and 5 mm laterally to the midline	Superficial	1	3 IU/7.5 SU	Avoid injecting into the DLI muscle
Masseter	The injection site is located between the mandibular angle and the facial pedicle	Deep	3	12 IU/30 SU	The anatomical danger is the facial pedicle
Platysma	In the mandibular region, the first injection point is located 10–15 mm posterior to the mandibular ligament. In case the second injection point is indicated, place it 10 mm posterior to the first	Superficial	2	4 IU/10 SU	The anatomical danger is the facial pedicle
	In the anterior neck, vertical series of injections into the platysma bands		6		Avoid injecting under the level of the thyroid cartilage
	In the posterior neck, the injection points are placed vertically from the mandibular border down to the thyroid cartilage level				

IU, international units; SU, Speywood units.

from the submental region to the neck with an interval of 1.0–1.5 cm, and in the posterior submandibular region. The tip of the needle penetrates into the skin and subcutaneous tissue, where the anterior border of the muscle is generally thin. The anterior band is held between the thumb and the index of the nondominant hand; the tip of the needle is placed 3–4 mm deep through the thin subcutaneous tissue and into the muscle.[9]

Downturned oral commissures caused by the DAO contraction may also require administering botulinum toxin in the mandibular region of the platysma muscle.

Injection is performed superficially. Never inject below the level of the thyroid cartilage since the platysma muscle is a thin muscle, and below the thyroid cartilage the toxin may spread into the neighboring peripharyngeal muscles and contribute to dysphonia and/or dysphagia.

Botulinum toxin injections in this muscle will produce relaxation of the platysma bands, improve the mandibular line, and decrease the horizontal neck lines.[27]

Copyright © 2015 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

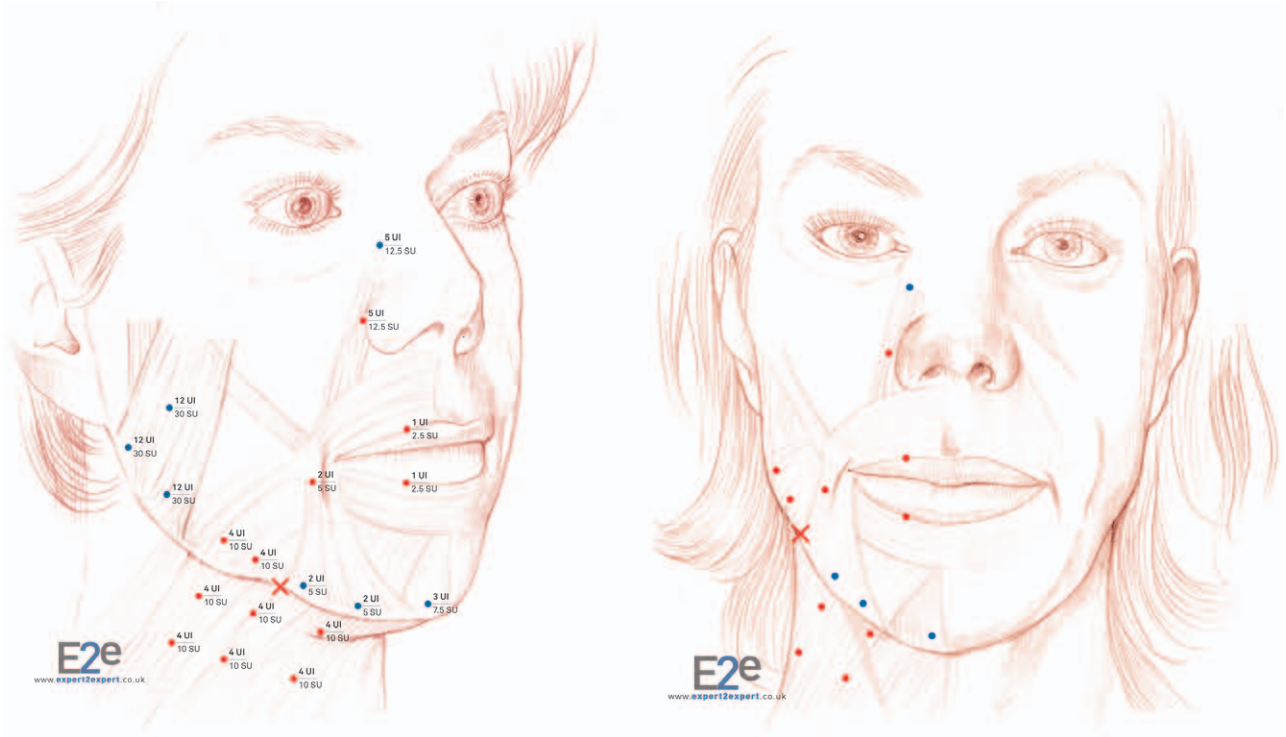


Fig. 6. Lower face injection points. The deep injection points in blue and the superficial injection points in red. The red cross shows the location of the facial pedicle on the mandible. IU indicates international units; SU, Speywood units. Figure courtesy of Expert2Expert-E2e.

Dose: 20–50 IU or 50–125 SU per side.[28]

Note that efficacy of botulinum toxin is related to the number of units and related to the muscular mass. In our cadaver studies, the muscular mass of the platysma is 5 times greater than the mass of the corrugator (Table 2 and Fig. 6).

CONCLUSIONS

Injecting botulinum toxin in the lower face can reverse signs of aging[29] and improve unsightly muscle movements and correct dynamic facial asymmetry. A detailed knowledge of the anatomy of the lower face is the key to a reliable botulinum toxin treatment.

Patrick Trévidic, MD
Expert2Expert Group
6 Rue Alfreed Roll
75017 Paris, France
patrick.trevidic@orange.fr

REFERENCES

1. Kane MA. The functional anatomy of the lower face as it applies to rejuvenation via chemodenervation. *Facial Plast Surg.* 2005;21:55–64.
2. Azib N, Charrier JB, Cornette de Saint Cyr B, et al. *Anatomy and Lip Enhancement.* Paris: E2e Medical Publishing/Master Collection 4; 2011.
3. Carruthers J, Carruthers A. Aesthetic botulinum A toxin in the mid and lower face and neck. *Dermatol Surg.* 2003;29:468–476.
4. Ingallina F, Trévidic P. *Anatomy and Botulinum Toxin Injections.* Paris: E2e Medical Publishing/Master Collection 1; 2013.
5. Criollo-Lamilla G, Ingallina F, Trévidic P. *Anatomy and Botulinum Toxin Injections.* 2nd ed. Paris: E2e Medical Publishing/Master Collection 1; 2015.
6. André P, Azib N, Berros P, et al. *Anatomy and Volumising Injections.* Paris: E2e Medical Publishing/Master Collection 2; 2012.
7. Loeb R. Naso-jugal groove leveling with fat tissue. *Clin Plast Surg.* 1993;20:393–400; discussion 401.
8. Shetty MK; IADVL Dermatosurgery Task Force. Guidelines on the use of botulinum toxin type A. *Indian J Dermatol Venereol Leprol.* 2008;74(Suppl):S13–S22.
9. Carruthers J, Fagien S, Matarasso SL; Botox Consensus Group. Consensus recommendations on the use of botulinum toxin type A in facial aesthetics. *Plast Reconstr Surg.* 2004;114(6 Suppl):1S–22S.
10. Tamura BM, Odo MY, Chang B, et al. Treatment of nasal wrinkles with botulinum toxin. *Dermatol Surg.* 2005;31:271–275.
11. Sucupira E, Abramovitz A. A simplified method for smile enhancement: botulinum toxin injection for gummy smile. *Plast Reconstr Surg.* 2012;130:726–728.
12. Revol M, Binder JP, Danino A, et al. *Manuel de chirurgie plastique, reconstructrice et esthétique.* 2ème éd. Paris: Editions Sauramps médical; 2009.

Copyright © 2015 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

13. Rogers CR, Mooney MP, Smith TD, et al. Comparative microanatomy of the orbicularis oris muscle between chimpanzees and humans: evolutionary divergence of lip function. *J Anat*. 2009;214:36–44.
14. Standring S. *Gray's Anatomy: the Anatomical Basis of Clinical Practice*. 39th ed. London: Churchill Livingstone; 2004.
15. Lightoller GH. Facial muscles: the modiolus and muscles surrounding the rima oris with some remarks about the panniculus adiposus. *J Anat*. 1925;60(Pt 1):1–85.
16. Lapatki BG, Oostenveld R, Van Dijk JP, et al. Topographical characteristics of motor units of the lower facial musculature revealed by means of high-density surface EMG. *J Neurophysiol*. 2006;95:342–354.
17. Gassia V, Beylot C, Béchaux S, et al. [Botulinum toxin injection techniques in the lower third and middle of the face, the neck and the décolleté: the "Nefertiti lift"]. *Ann Dermatol Venereol*. 2009;136(Suppl 4):S111–S118.
18. Carruthers JD, Glogau RG, Blitzer A; Facial Aesthetics Consensus Group Faculty. Advances in facial rejuvenation: botulinum toxin type a, hyaluronic acid dermal fillers, and combination therapies–consensus recommendations. *Plast Reconstr Surg*. 2008;121(5 Suppl):5S–30S; quiz 31S.
19. Rohrich RJ. Advances in facial rejuvenation: botulinum toxin A. Hyaluronic acid dermal fillers, and combination therapies–consensus recommendations. Editor's Foreword. *Plast Reconstr Surg*. 2008;121(5 Suppl):1S–2S.
20. Carruthers A, Carruthers J, Monheit GD, et al. Multicenter, randomized, parallel-group study of the safety and effectiveness of onabotulinumtoxinA and hyaluronic acid dermal fillers (24-mg/ml smooth, cohesive gel) alone and in combination for lower facial rejuvenation. *Dermatol Surg*. 2010;36(Suppl 4):2121–2134.
21. Gray's Anatomy [Internet]. New York: the muscles of mastication. Available at: http://web.archive.org/web/20041209015011/http://education.yahoo.com/reference/gray/subjects/subject?id=109. Accessed April 3, 2015.
22. Kahn JL, Bourjat P. Le Masséter: Un muscle énigmatique. *J Radiol* 2001;82:547–552.
23. Iwasaki S, Tokunaga T, Baba S, et al. Noninvasive estimation of the location of the end plate in the human masseter muscle using surface electromyograms with an electrode array. *J Osaka Dent Univ*. 1990;24:135–140.
24. Mito K, Sakamoto K. Distribution of muscle fiber conduction velocity of m. masseter during voluntary isometric contraction. *Electromyogr Clin Neurophysiol*. 2000;40:275–285.
25. Goldman A, Wollina U. Elevation of the corner of the mouth using botulinum toxin type A. *J Cutan Aesthet Surg*. 2010;3:145–150.
26. Prager W, Bee EK, Havermann I, et al. IncobotulinumtoxinA for the treatment of platysmal bands: a single-arm, prospective proof-of-concept clinical study. *Dermatol Surg*. 2015;41(Suppl 1):S88–S92.
27. Kane MA. Nonsurgical treatment of platysmal bands with injection of botulinum toxin A. *Plast Reconstr Surg*. 1999;103:656–663; discussion 664.
28. Matarasso A, Matarasso SL, Brandt FS, et al. Botulinum A exotoxin for the management of platysma bands. *Plast Reconstr Surg*. 1999;103:645–652; discussion 653.
29. Guyuron B, Huddleston SW. Aesthetic indications for botulinum toxin injection. *Plast Reconstr Surg*. 1994;93:913–918.

Copyright © 2015 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.